DATA ANALYST PORTFOLIO PROJECT

COFFEE SHOP ANALYTICS

From Raw Data to Business Strategy

CSV FILES → POSTGRESQL → POWER BI

Asghar Hussain • Data Analyst • June 2026

01 — THE CHALLENGE

The Business Problem

Every coffee shop owner faces the same question:

"How do we increase revenue without losing customers?"

The client needed data-backed answers to three critical questions:

1

Should we increase prices?

And on which specific products — without hurting demand?

2

Who are our most valuable customers?

And what will it take to keep them coming back?

3

Which categories deserve investment?

And which should be phased out to protect margin?

02 — PROJECT SCOPE

The Data Behind the Story

450+	50	~2,000	3
Customers	Products across 5 categories	Orders analyzed	Years of data (2023–2025)

THE PIPELINE

3 CSV Files
Customers, products & orders

→

PostgreSQL
Normalized schema + advanced views

→

Power BI
6-page interactive dashboard

Every insight in this deck traces back to this pipeline — built and validated end-to-end.

03 — ARCHITECTURE

How I Built It

PHASE 1

PostgreSQL — Data Loading & Views

- Normalized tables: customers, products, orders
- COPY commands to import 3 years of CSV data
- 5 advanced SQL views built with CTEs, window functions & self-joins
- RFM segmentation, price simulation, BCG matrix, cohort retention, category trends

→

PHASE 2

Power BI — Dashboard & UX

- 6 interactive pages, one consistent theme
- Conditional formatting: data bars & color scales
- DAX measures for KPIs, YoY growth & what-if parameters
- Tooltips, slicers, and reset buttons throughout

→

PHASE 3

AI-Assisted Development

- Claude AI for SQL optimization, DAX generation & documentation
- DeepSeek for troubleshooting & code refinement
- ~40% reduction in overall development time

Six Pages, One Story

01

Executive Summary

Overall business health

KPI cards, forecast, top 5 products

02

Customer Insights

Customer segmentation

RFM table, heatmap, decomposition tree

03

Pricing Strategy

Price increase simulation

KPI cards, product table, scatter plot

04

Portfolio (BCG)

BCG matrix strategy

BCG scatter plot, category table

05

Retention Analysis

Customer loyalty analysis

Cohort matrix, trend line chart

06

Category Trends

Year-over-year revenue

Line chart, data-bar table

Executive Summary — The 30,000-Foot View

Live Power BI dashboard — Executive Summary page

$871.33K	$474.53K	54%
TOTAL REVENUE	TOTAL PROFIT	PROFIT MARGIN

KEY FINDINGS

- Total revenue of $871.33K at a healthy 54% profit margin; average order value of $196
- West region leads with 42% of total revenue — the strongest market to nurture
- Grinders & Brewers is the top category at 58% of revenue and 62% of profit
- YoY revenue growth is down 18.58% — a signal worth investigating further

KEY INSIGHT

"Grinders & Brewers dominates revenue and profit — a clear Cash Cow to protect."

Customer Insights — Who Are Our Best Customers?

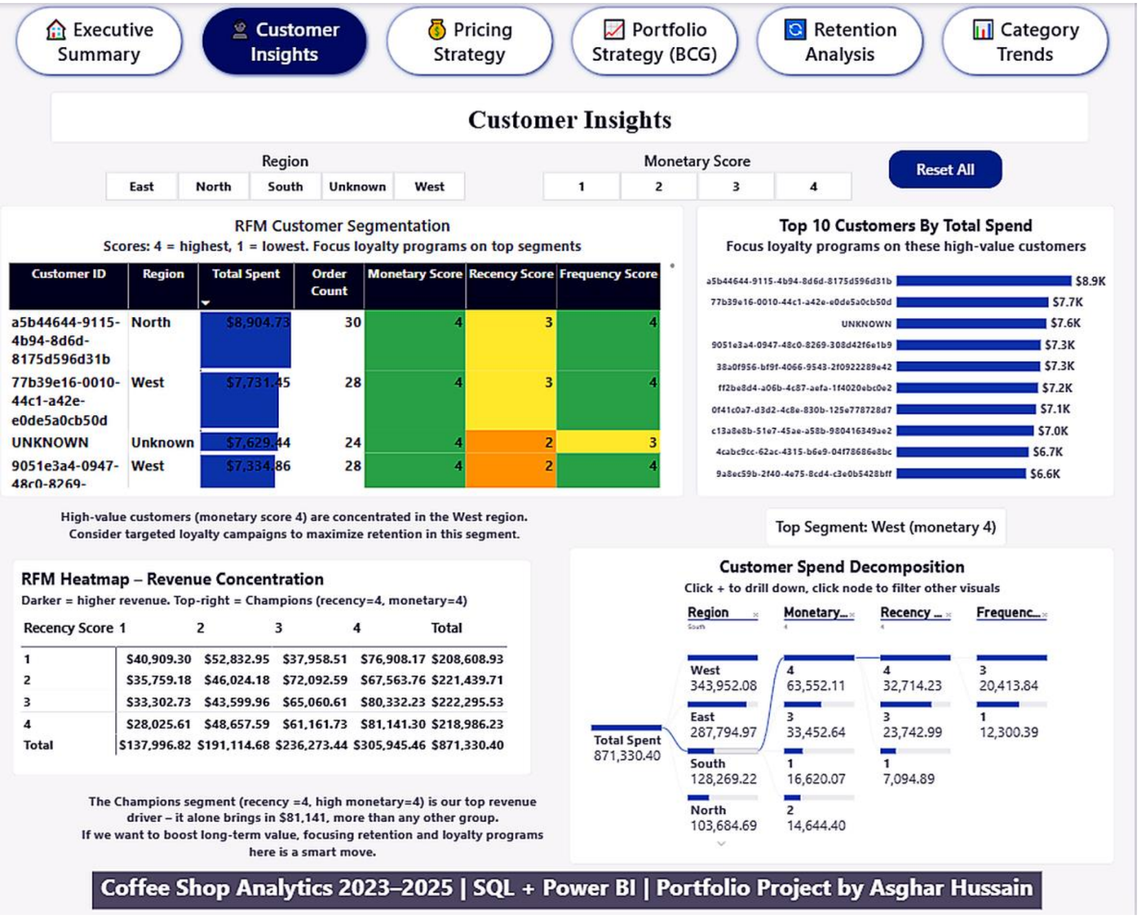
$96K	$81.1K	30
TOP-10 REVENUE	CHAMPIONS SEGMENT	TOP ORDERS/CUST.

KEY FINDINGS

- Top 10 customers contribute roughly $96K — about 11% of total revenue
- RFM “Champions” (recency = 4, monetary = 4) alone generate $81,141 — more than any other segment
- Top single customer spent $8,904.73 across 30 orders
- High-value customers (monetary score 4) are concentrated in the West region

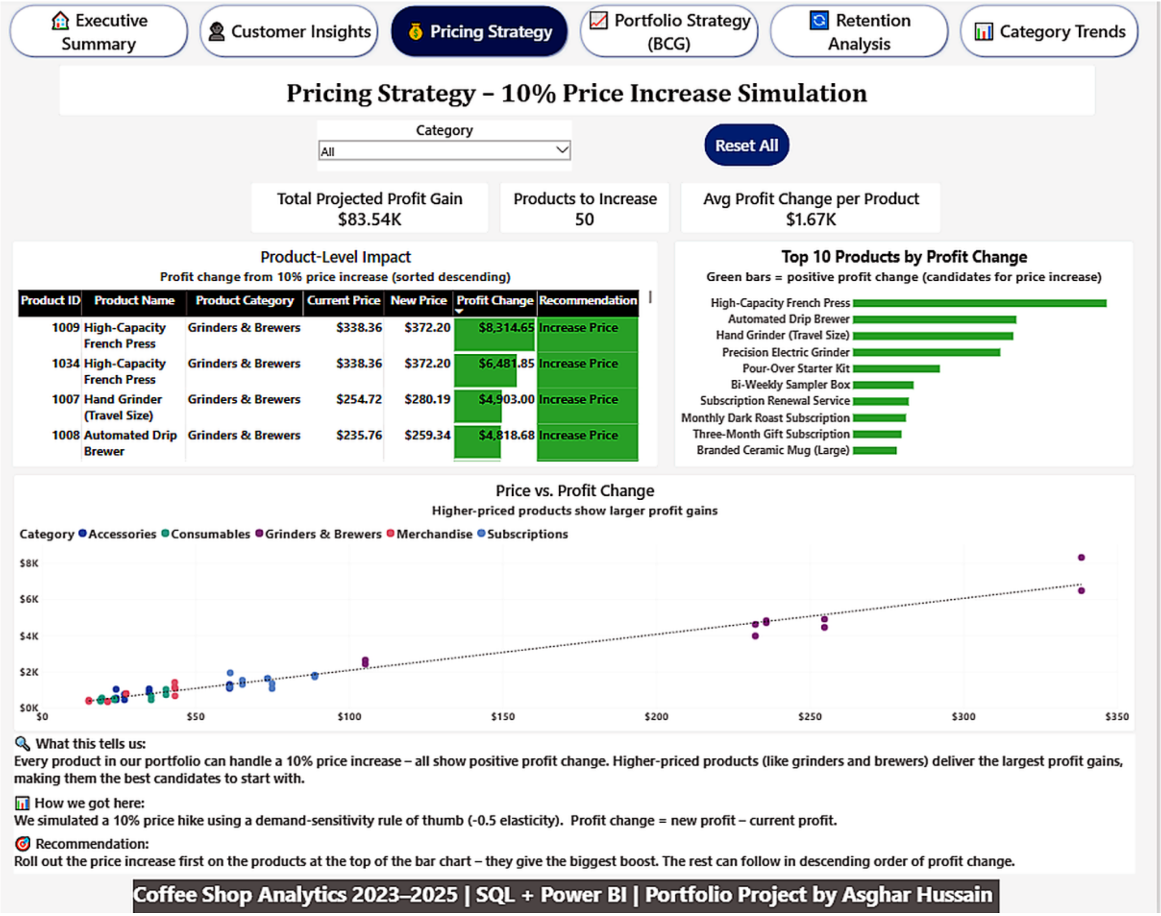
KEY INSIGHT

“Champions (top-right RFM cell) should anchor the new loyalty program.”



Live Power BI dashboard — Customer Insights page

Pricing Strategy — 10% Price Increase Simulation



Live Power BI dashboard — Pricing Strategy page

$83.54K
PROJECTED PROFIT GAIN

50
PRODUCTS TO INCREASE

$1.67K
AVG. GAIN / PRODUCT

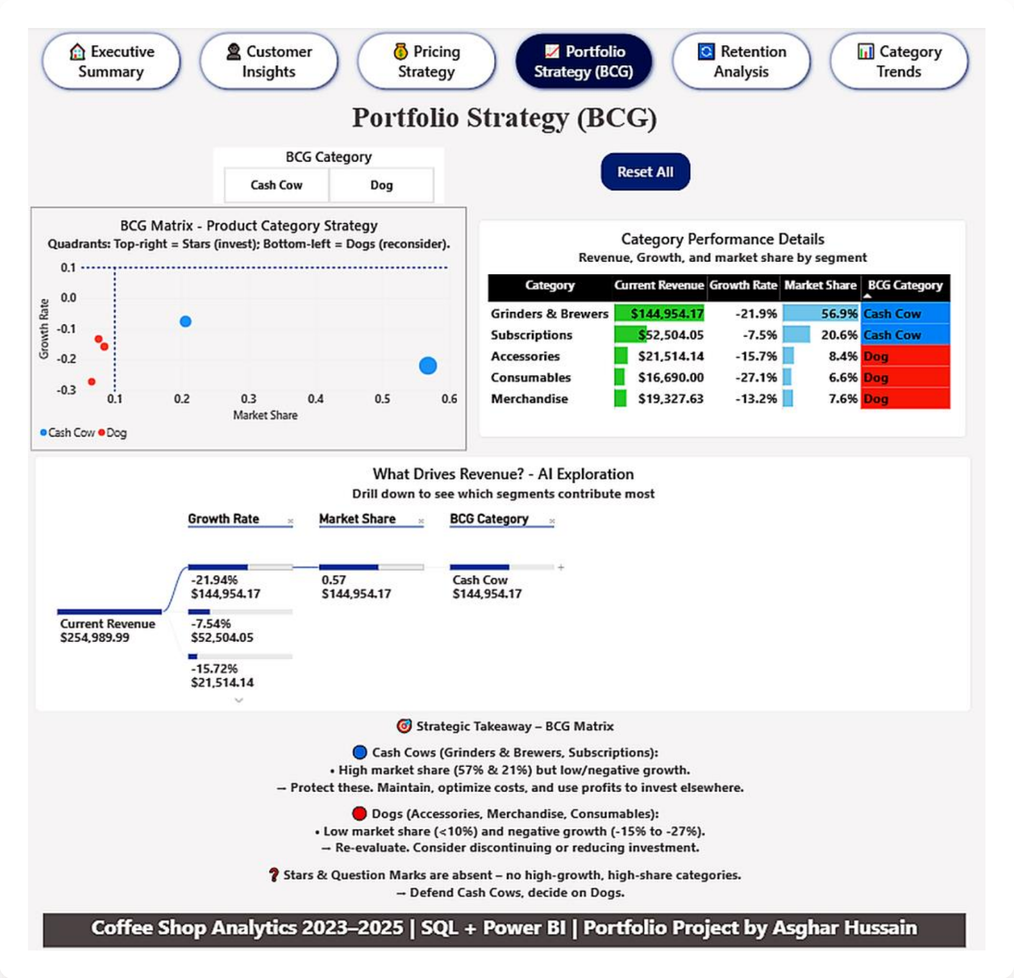
KEY FINDINGS

- Every product in the portfolio shows a positive profit change under a simulated 10% price increase
- Top gainer: High-Capacity French Press — +$8,314.65 in projected profit
- Hand Grinder (Travel Size) and Automated Drip Brewer follow closely, both over +$4,800
- Higher-priced Grinders & Brewers products deliver the largest absolute gains

KEY INSIGHT

"Grinders & Brewers products give the biggest profit boost — start the rollout there."

Portfolio Strategy — BCG Matrix



Live Power BI dashboard — Portfolio Strategy (BCG) page

Category	Growth	Share	BCG
Grinders & Brewers	-21.9%	56.9%	Cash Cow
Subscriptions	-7.5%	20.6%	Cash Cow
Accessories	-15.7%	8.4%	Dog
Consumables	-27.1%	6.6%	Dog
Merchandise	-13.2%	7.6%	Dog

KEY INSIGHT

- Grinders & Brewers and Subscriptions are Cash Cows — high share, but declining growth
- Accessories, Consumables & Merchandise are Dogs — low share and steep decline
- No Stars or Question Marks exist in the current portfolio

STRATEGIC TAKEAWAY

“Protect both Cash Cows. Re-evaluate the three Dogs for discontinuation or cost reduction.”

Retention Analysis — Do Customers Come Back?

100%
MONTH 1

~45%
MONTH 2

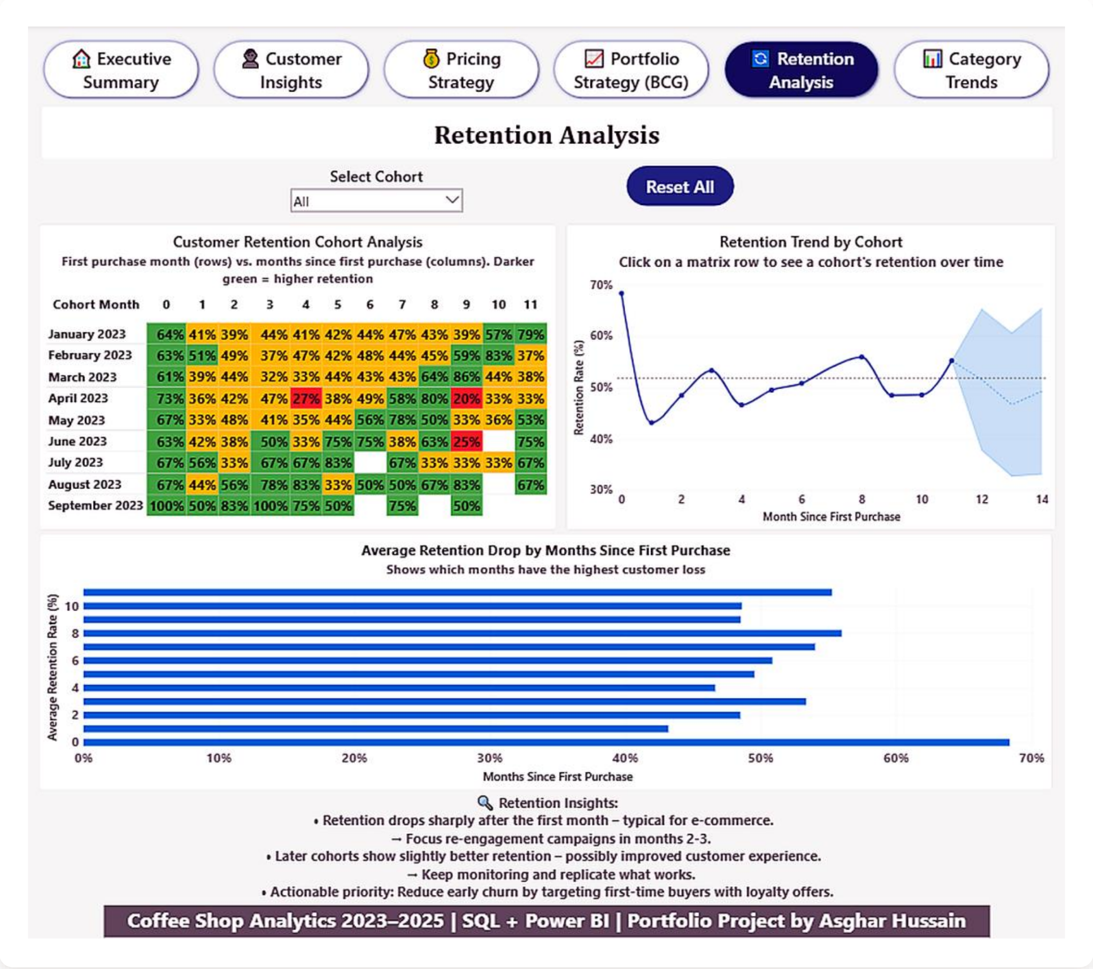
~33%
MONTH 3

KEY FINDINGS

- Retention falls by roughly 55 points between month 1 and month 2 — a sharp early drop-off
- Cohort matrix shows the steepest customer loss consistently happens right after the first purchase
- Later cohorts (2024–2025) show slightly better retention than earlier ones
- Pattern is typical for e-commerce, but still represents the single biggest opportunity to act on

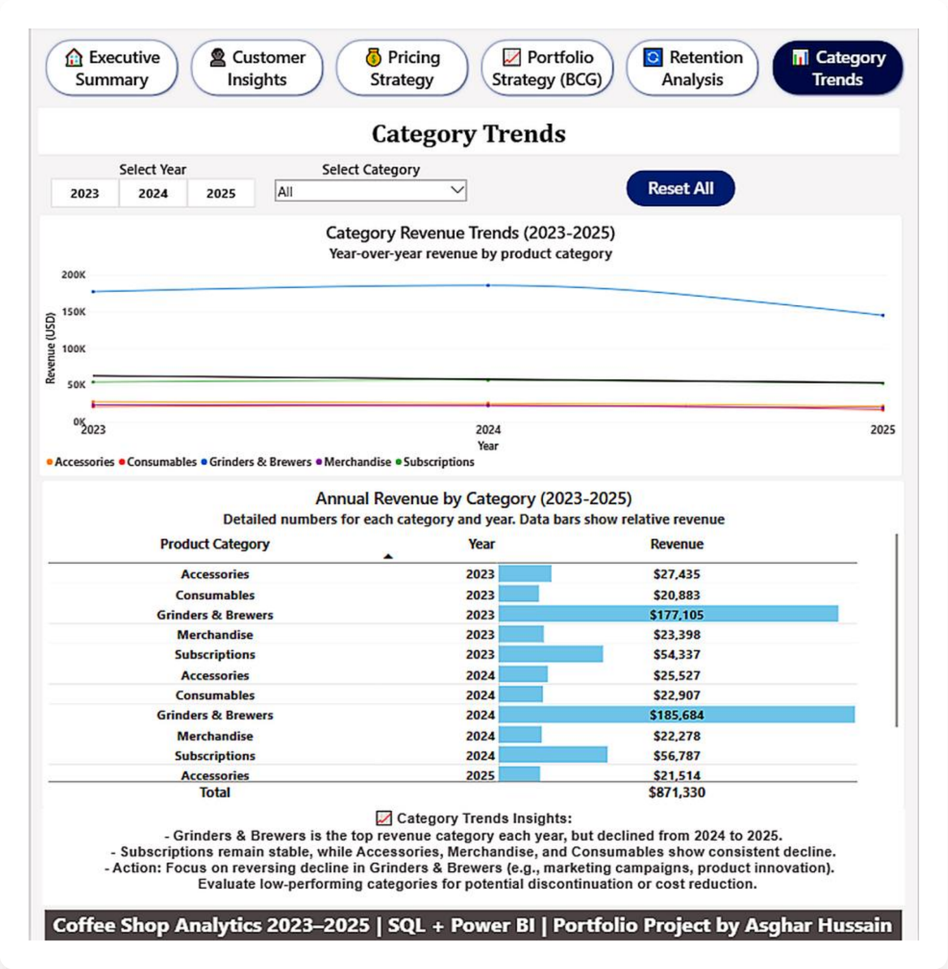
KEY INSIGHT

"Retention drops sharply after month 1 — focus re-engagement campaigns on months 2–3."



Live Power BI dashboard — Retention Analysis page

Category Trends — Year-over-Year Revenue



Live Power BI dashboard — Category Trends page

$871.3K

3-YR TOTAL REVENUE

$185.7K

G&B PEAK (2024)

KEY FINDINGS

- Grinders & Brewers is the top revenue category every year, but declined from 2024 to 2025
- Subscriptions remain the most stable category across all three years
- Accessories, Merchandise, and Consumables all show consistent year-over-year decline
- Recommended action: reinvest in marketing and product innovation for Grinders & Brewers

KEY INSIGHT

“Grinders & Brewers has been declining since 2024 — worth investigating the root cause.”

Key Business Insights & Recommended Actions

Insight	What the Data Says	Business Action
Pricing Power	10% price increase shows a positive profit impact on all 50 products (avg. +$1.67K each)	Implement in phases — start with Grinders & Brewers
Customer Concentration	Top 10 customers drive ~$96K (11%) of revenue; Champions segment alone drives $81.1K	Launch a loyalty program & premium subscriptions for top spenders
Cash Cows vs. Dogs	Grinders & Brewers + Subscriptions = Cash Cows (77.5% combined share); the rest are Dogs	Protect & optimize Cash Cows; re-evaluate Dogs for discontinuation
Retention Dip	Retention falls from 100% to roughly 45% by the second month	Focus re-engagement campaigns on customer months 2–3

Bottom line: protect what's working, price with confidence, and re-engage customers before they churn.

THANK YOU

Let's Build Something Data-Driven Together

I'm a data analyst who turns messy data into clear, actionable business insight. This project reflects my approach: business-first, data-driven, and human-centered.

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